How-To Library

Informatica

Setting Up Disaster Recovery for PowerCenter

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Abstract

You can implement a disaster recovery plan for PowerCenter deployments to recover from accidental loss of user data, hardware failures, database corruption, and natural disasters. This article presents some recovery strategies and best practices that you can adopt to recover and protect PowerCenter deployments during a disaster.

Supported Versions

- PowerCenter 9.x-10.0

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Overview

A typical disaster recovery plan includes PowerCenter installations at geographically different locations that work together to achieve recovery. To avoid a data outage that results from a disaster, you must frequently back up the contents of the PowerCenter repository so that you can restore it to a new database when the PowerCenter repository database fails.

This article describes how to set up a disaster recovery plan for the following scenarios:

- Disaster recovery for PowerCenter with the high availability option
- Disaster recovery for PowerCenter without the high availability option
- Disaster recovery when a backup of the domain configuration repository does not exist
- Disaster recovery in a clustered environment

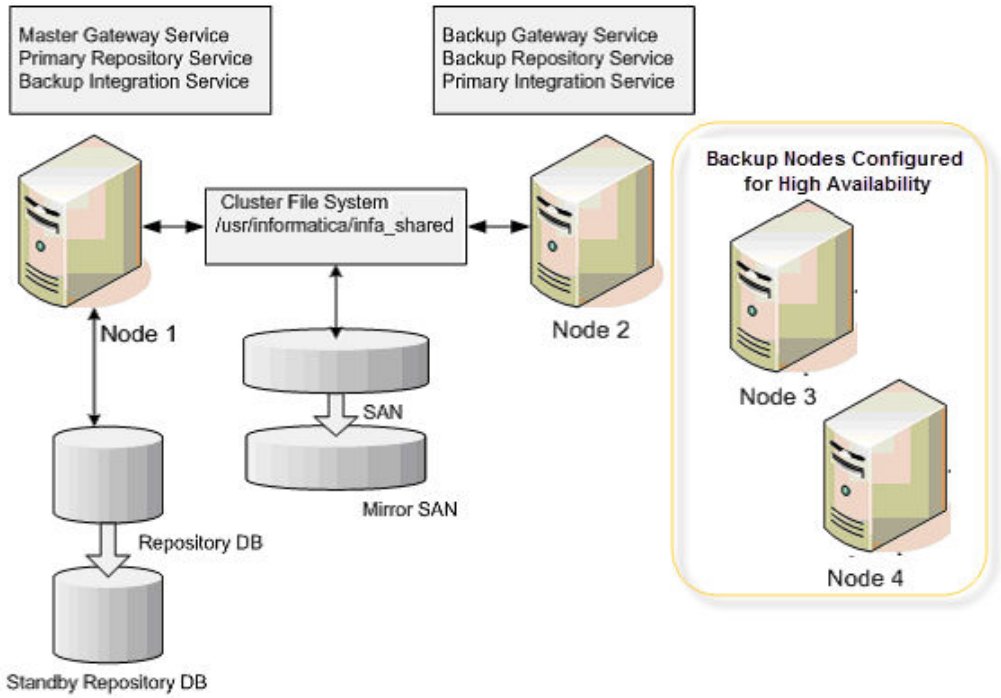
Disaster Recovery for PowerCenter with the High Availability Option

If you run critical and time-sensitive operational applications across multiple business and functional areas, set up a PowerCenter domain for high availability and disaster recovery.

Informatica recommends that you use the PowerCenter high availability option to leverage resources on all of the servers. You require a single cluster file system across all nodes to coordinate read or write access to the storage pool, ensure data integrity, and attain performance.

For example, your organization has installed PowerCenter on four nodes and configured the nodes for disaster recovery.

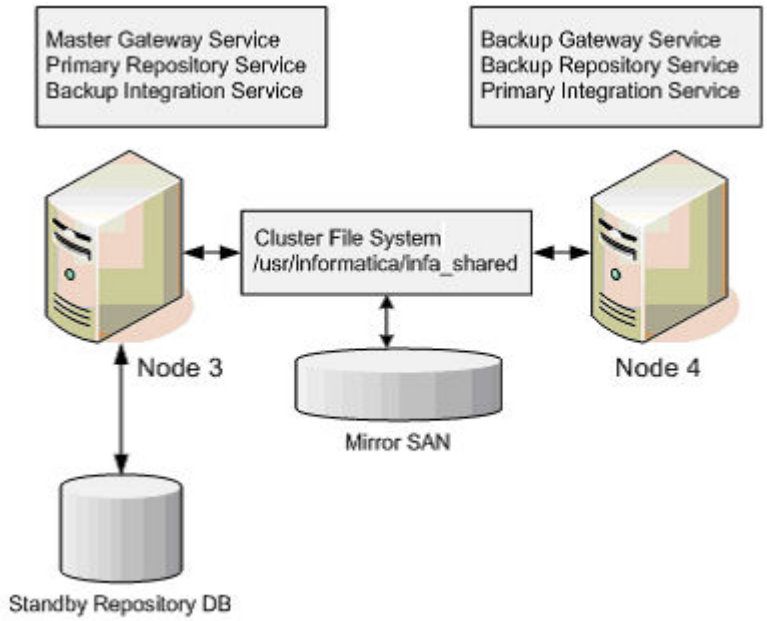
The following image shows the primary data center during a normal operation:



Each node runs a Service Manager process that controls the services running on that node. During a normal operation, the Service Manager runs on Node1 and Node2. Node3 and Node4 are backup nodes configured for high availability and are not available during a normal operation.

A node can become unavailable if the Service Manager fails or if the administrator shuts down the machine or the Service Manager process. The Service Manager processes periodically exchange a heartbeat signal among themselves to detect any node or network failure. When the Service Manager process detects a primary or backup node failure, the remaining nodes determine the new primary or backup node through a distributed voting algorithm.

The following image shows the backup data center after failover from the primary data center:



When the Service Manager fails, Node1 and Node2 are unavailable. Start Node3 and Node4 in the backup data center. Node3 and Node4 try to establish a connection with the master gateway Node1. When the nodes fail to reach Node1, Node3 or Node4 becomes the new master gateway.

When normal operations resume, Node1 and Node2 restart and the Informatica Service Manager process starts on these nodes. As the Informatica Service Manager process is shutdown on Node3 and Node4, Node1 tries to become the master gateway and succeeds.

Typically, the administrator configures the operating system to start the Service Manager when the operating system starts or when the Service Manager fails unexpectedly. As the Service Manager is the primary point of control for PowerCenter services on a node, you must use monitoring scripts to detect unexpected failures of the Service Manager. To automate the switchover to disaster recovery, you can set the change in configuration required for the disaster recovery servers as a script. For example, configure the database connectivity such that failover to the standby database occurs without manual intervention. All database connectivity information must be identical in both data centers to use the same source and target databases.

Note: If any of the scheduling tools, FTP servers, and message queues connect to PowerCenter, set these applications to connect to PowerCenter in the backup data center.

Prerequisites

You must complete the prerequisites before you can set up disaster recovery for PowerCenter with the high availability option.

- Verify that you have the following resources:
  - PowerCenter high availability option license.
  - Shared file system that Informatica supports for concurrent file access from multiple servers.
  - Geographically distributed machines to install the PowerCenter components.
  - Shared file system that Informatica supports for concurrent file access from multiple servers.

- A standby database server that contains replicated logs for recovery of the database tier. The server must provide disaster tolerance for both the PowerCenter repository and the domain database. The standby database server must have performance characteristics in parity with the primary PowerCenter repository database server.

- If you want to reduce the recovery time for storage, you can use near real-time replication of data from the primary SAN to a mirror SAN . Consult storage vendors for optimal SAN and mirror SAN configuration.

Setting Up Disaster Recovery for PowerCenter by Using High Availability

If your license includes the high availability option, install and configure PowerCenter on multiple nodes to enable disaster recovery.

For example, install PowerCenter on four nodes and then configure the nodes for disaster recovery. Perform the following tasks:

1. When you install PowerCenter on Node1, select the **Create Domain** option.

   Node1 becomes the master gateway.
2. For Node2, Node3, and Node4 installations, select the **Join the Domain** option.
3. For Node2, Node3, and Node4, select the **Serves as Gateway** option.
4. Verify that Node1 is the Master Gateway.

   You can use the following command to ping the master node for verification: `infacmd Ping`

Setting Up Disaster Recovery for PowerCenter Components by Using High Availability

Configure users, user access, services, and the shared directory for disaster recovery.

For example, you have installed PowerCenter on four nodes and then you want to set up the nodes for disaster recovery.

To set up the PowerCenter nodes for disaster recovery, perform the following tasks:

1. Create the same user ID and group ID for users on all the nodes.
2. Place the infa_shared directory on a shared file system, such as Common File System (CFS), and grant all users with read or write access to the files. For example, if a workflow that runs on Node1 creates a log file in the log directory, the rest of the nodes must be able to read and update this file.
3. Set up the PowerCenter services for disaster recovery:
   a. Create the PowerCenter Repository Service, and select Node1 as the primary node.
   b. Create the PowerCenter Integration Service, and select Node2 as the primary node.
   c. Assign Node3 and Node4 as backup nodes for the PowerCenter Repository Service and PowerCenter Integration Service.

      Note: The backup nodes must have access to the standby PowerCenter repository database.
4. Configure the persistent cache files, parameter files, logs, and other run-time files to use the directory created on the shared file system, and point the $PMRootDir variable to this directory.

   You can choose to create a symbolic link from the default infa_shared location to the infa_shared directory created on the shared file system.

For example, if you have installed PowerCenter on a UNIX operating system, perform the following steps:

a. Create the directory an any location on UNIX, such as `/home/<username>/infa_shared`, provided that all the hosts can access this location.

   Each installation of the node has the following default infa_shared directory: `$INFA_HOME/server/infa_shared`

b. Remove the infa_shared directory, and then run the following command to create a link to the shared location: `ln -fs /home/<username>/infa_shared infa_shared`

c. Verify that the shared infa_shared structure that contains the directories is the same as the default structure.

   Note: If you configure the PowerCenter Integration Service to run on a grid, you do not need to change the $PMRootDir for the individual service processes.

Disaster Recovery for PowerCenter Without the High Availability Option

If your license does not include the high availability option, you must maintain a second installation of PowerCenter and a database server to contain the domain configuration repository. You can also choose to have a separate set of PowerCenter binaries and a database server that contains the PowerCenter and domain configuration repository.

You must periodically back up the PowerCenter repository, domain configuration repository, users, and groups. When the system shuts down unexpectedly, bring up the PowerCenter domain on the second machine and restore the working status.

Setting up Disaster Recovery for PowerCenter Components Without Using High Availability

Perform the following tasks for the PowerCenter components installed on the first machine:

1. To back up the domain configuration repository from the first machine to an `.mrep` file, run the following command: `infasetup backupDomain`
2. To back up the PowerCenter repository to an mrep file, run the following command: `pmrep backup`
3. To back up and export users and groups to an XML file, run the following command: `infacmd exportUsersAndGroups`

Perform the following tasks to recover the PowerCenter components on the second machine:

1. Restore the backed up mrep file to the second database with the -force option. To restore the domain configuration repository, run the following command: `infasetup RestoreDomain`
2. Start the disaster recovery node.
3. To import users and groups to the domain from the XML file, run the following command: `infacmd importUsersAndGroups`
4. To import and restore the PowerCenter repository, run the following command: `pmrep restore`

For more information about the commands, see the *Informatica Command Reference*.

Disaster Recovery When a Backup of the Domain Configuration Repository Does Not Exist

Consider a scenario where you want to recover a PowerCenter domain configuration repository where the database backup is available but a backup of the file structure is not available.

If a backup of the domain configuration repository files does not exist, you cannot create a new domain and you cannot join an existing domain. In this scenario, you require a new PowerCenter installation, which creates the domain database tables. If the tables exist in the current user schema, you need to configure a domain for another user. Because you do not want to create another domain but only extract the file structure, you need to stop the installation after the binaries are extracted.

Configuring the PowerCenter Installation for Disaster Recovery

After you create the file structure in the new installation, you must manually proceed with the following tasks:

1. Back up the domain from the existing domain configuration database tables. Run the following command:
   - For Windows: `infasetup.bat backupdomain`
   - For UNIX: `infasetup.sh backupdomain`
     Provide the same domain name of the PowerCenter domain that you want to recover.
2. Copy the backup file to the new machine where you want to install Informatica.

   Note: You must cancel the installation before configuring the domain. To cancel, click **Cancel** in graphical mode and **CTRL+C** in console mode when the installer prompts you to create or join the domain.
3. Restore the domain that you backed up. Run the following command:
   - For Windows: `infasetup.bat restoredomain`
   - For UNIX: `infasetup.sh restoredomain`

   Ensure that you do not use the `-ca` flag. The `-ca` flag clears all node associations.
4. Create the `nodemeta.xml` file that contains the configurations of the node. Run the following command:
   - For Windows: `infasetup.bat definegatewaynode`
   - For UNIX: `infasetup.sh definegatewaynode`

   This command overwrites the `nodemeta.xml` file that stores the configuration metadata for the node. The node uses the `nodemeta.xml` to recognize where the domain database is present and the node name it uses to start.
5. Start the node. Use the following command:
   - For Windows: `infaservice.bat startup`
   - For UNIX: `infaservice.sh startup`

   For more information about the commands, see the *Informatica Command Reference*.

Disaster Recovery in a Clustered Environment

You can configure disaster recovery for PowerCenter in a clustered environment.

Consider two systems Production_HQ and Production_CA that are running as a cluster. The PowerCenter repository services are running on both systems. When the Production_CA machine shuts down unexpectedly, the Production_HQ machine must take over.

Prerequisites

Before you begin, consider the following prerequisites:

- Each of the machines must have multiple network cards.
- The cluster must be accessible when you use a physical IP address. Each machine in the cluster must be accessible when you use a physical IP address. Each part of the cluster must also have one or more logical IP addresses. The logical IP addresses move to the other machine at failover time.

Sample Configuration of Production_CA and Production_HQ Machine

The Production_CA machine contains the following configurations:

- PowerCenter Repository Service listens on port 5001.
- Repository Agents run on ports 5002-5005.
- Informatica services listen on port 4001.
- The IP address is a logical IP address of the Production_CA machine.

The Production_HQ machine contains the following normal and failover configurations:

Normal

The normal machine contains the following configurations:

- PowerCenter Repository Service listens on port 5011.
- Repository Agents run on ports 5012-5015.

  Note: Ensure that the PowerCenter Repository Service and Repository Agents run on different ports to avoid conflicts during failover.
- Informatica services listen on port 4011.
- The IP address can either be a logical or physical IP address of the Production_HQ machine.

Failover

The failover machine takes over the Production_CA machine and contains the following configurations:

- PowerCenter Repository Service listens on port 5001.
- Repository Agents run on ports 5002-5005.
- Informatica services listen on port 4001.
- The IP address used is the logical IP address of the Production_CA machine, as the IP address swaps to this machine at failover time.

Disaster Recovery Examples for Business Continuity

Consider the following external dependencies while you plan for disaster recovery of the Informatica services:

- Informatica shared file system synchronization between the primary site and disaster recovery site.
- Database synchronization of the domain database and the repository databases between the primary site and the disaster recovery site.
- Virtual IP failover from the primary to the disaster recovery site.
- Source systems and target systems failover to their respective disaster recovery sites.

You must design an enterprise-level disaster recovery solution based on the acceptable downtime for the business processes when a disaster happens.

If you have a solution for synchronization of the file system and database, you could consider the following examples for disaster recovery.

Example 1

The primary Informatica services are in California. The New Jersey servers serve as a backup for the Informatica services for disaster recovery:

Installation Details

Install PowerCenter only on primary nodes by using the virtual host name during the domain creation.

High-Level Steps After a Disaster With Virtual IP Failover

Perform the following steps:

1. Fail over the virtual IP.
2. Synchronize the file system at the Informatica installation location and the infa_shared run-time file location.
3. Manually start the nodes in the New Jersey server and start the jobs from the failure point.

Example 2

The primary Informatica services are in California. The New Jersey servers host other applications and serve as a backup for the Informatica services for disaster recovery.

Installation Details

Perform the following steps:

- Install Informatica PowerCenter on the disaster recovery nodes when you install the primary nodes. The nodes remain down unless you manually start the nodes as part of the disaster recovery process.
- Assign one of the nodes in the disaster recovery site as the Gateway node.

High-Level Steps After a Disaster Without Virtual IP Failover

Perform the following steps:

1. Synchronize the infa_shared file system in the run-time file location.
2. Manually start the nodes in the New Jersey server and start the jobs from the failure point.

Example 3

There are two primary Informatica domains: A New Jersey domain with two nodes and a California domain with two nodes. In a disaster, the nodes in the California domain fail over to the New Jersey site.

Installation Details

Install PowerCenter on all the four nodes by using the virtual host name during the domain creation step in the installation process.

High-Level Steps After a Disaster With Virtual IP Failover

Perform the following steps:

1. Fail over the virtual IP.
2. Synchronize the file systems both at the Informatica installation location and the infa_shared run-time file location.
3. Manually bring up the nodes in the New Jersey server and start the jobs from the failure point.

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